



MANIPAL

ACADEMY of HIGHER EDUCATION

(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5103 Linux OS and Scripting – 2 Theory

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22/08/2026



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Compare Permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

1. Setting up:

```
(kali㉿kali)-[~]
$ touch sample.txt

(kali㉿kali)-[~]
$ mkdir sample_dir

(kali㉿kali)-[~]
$ touch sample_dir/aansh1.txt

(kali㉿kali)-[~]
$ ls -l
total 52
drwxrwxr-x 3 kali kali 4096 Aug 13 20:07 aansh
-rw-rw-r-- 1 kali kali 0 Aug 20 13:42 access.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:24 admin.html
-rw-rw-r-- 1 kali kali 0 Aug 20 12:44 a.txt
-rw-rw-r-- 1 kali kali 0 Aug 20 13:42 auth.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:42 auth.log.1
-rw-rw-r-- 1 kali kali 0 Aug 20 13:42 auth.log.2
-rw-rw-r-- 1 kali kali 0 Aug 20 13:24 backup1.html
-rw-rw-r-- 1 kali kali 0 Aug 20 13:24 backup2.txt
-rw-rw-r-- 1 kali kali 0 Aug 20 12:47 b.txt
drwxr-xr-x 2 kali kali 4096 Jul 30 12:34 Desktop
drwxr-xr-x 2 kali kali 4096 Jul 30 12:44 Documents
drwxr-xr-x 3 kali kali 4096 Aug 11 00:55 Downloads
-rw-rw-r-- 1 kali kali 0 Aug 20 13:42 error.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:48 failed_login.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:24 file123.c
-rw-rw-r-- 1 kali kali 0 Aug 20 13:48 firewall.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:24 index.html
drwxrwxr-x 2 kali kali 4096 Aug 20 13:07 kali
drwxrwxr-x 2 kali kali 4096 Aug 13 19:06 LinuxLab
-rw-rw-r-- 1 kali kali 0 Aug 20 13:48 login_backup.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:24 login.html
-rw-rw-r-- 1 kali kali 0 Aug 20 13:48 login.txt
-rw-rw-r-- 1 kali kali 0 Aug 20 13:47 malware1.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:47 malware2.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:47 malwareA.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:47 malware_report.txt
drwxr-xr-x 2 kali kali 4096 Jul 30 12:34 Music
-rw-rw-r-- 1 kali kali 0 Aug 20 13:47 normal.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:42 notes.txt
drwxr-xr-x 2 kali kali 4096 Jul 30 12:34 Pictures
drwxr-xr-x 2 kali kali 4096 Jul 30 12:34 Projects
drwxr-xr-x 2 kali kali 4096 Jul 30 12:34 Public
-rw-rw-r-- 1 kali kali 0 Aug 20 13:24 report.txt
drwxrwxr-x 2 kali kali 4096 Aug 22 13:15 sample_dir
-rw-rw-r-- 1 kali kali 0 Aug 22 13:14 sample.txt
-rw-rw-r-- 1 kali kali 0 Aug 20 13:48 successful_login.log
-rw-rw-r-- 1 kali kali 0 Aug 20 13:42 system.log
```

2. 400(r---)

```
(kali㉿kali)-[~]
$ chmod 200 sample.txt

(kali㉿kali)-[~]
$ chmod 200 sample_dir

(kali㉿kali)-[~]
$ ls -ld sample.txt sample_dir
d-w----- 2 kali kali 4096 Aug 22 13:15 sample_dir
--w----- 1 kali kali    0 Aug 22 13:14 sample.txt

(kali㉿kali)-[~]
$ cat sample
cat: sample: No such file or directory

(kali㉿kali)-[~]
$ echo "Hello" > sample.txt

(kali㉿kali)-[~]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(kali㉿kali)-[~]
$ cd sample_dir
cd: permission denied: sample_dir

(kali㉿kali)-[~]
$ touch sample_dir/aansh1.txt
touch: cannot touch 'sample_dir/aansh1.txt': Permission denied
```

3. 200(-w-)

```
(kali㉿kali)-[~]
$ chmod 400 sample.txt

(kali㉿kali)-[~]
$ chmod 400 sample_dir

(kali㉿kali)-[~]
$ ls -ld sample.txt sample_dir
dr----- 2 kali kali 4096 Aug 22 13:15 sample_dir
-r----- 1 kali kali    0 Aug 22 13:14 sample.txt

(kali㉿kali)-[~]
$ cat sample.txt

(kali㉿kali)-[~]
$ echo "Hello" > sample.txt
zsh: permission denied: sample.txt

(kali㉿kali)-[~]
$ ls sample_dir
ls: cannot access 'sample_dir/aansh1.txt': Permission denied
aansh1.txt

(kali㉿kali)-[~]
$ cd sample_dir
cd: permission denied: sample_dir
```

4. 100(--x)

```
(kali㉿kali)-[~]
└─$ chmod 100 sample.txt

(kali㉿kali)-[~]
└─$ chmod 100 sample_dir

(kali㉿kali)-[~]
└─$ ls -ld sample.txt sample_dir
d--x----- 2 kali kali 4096 Aug 22 13:15 sample_dir
--x----- 1 kali kali    6 Aug 22 13:27 sample.txt

(kali㉿kali)-[~]
└─$ cat sample.txt
cat: sample.txt: Permission denied

(kali㉿kali)-[~]
└─$ ./sample.txt
zsh: permission denied: ./sample.txt

(kali㉿kali)-[~]
└─$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(kali㉿kali)-[~]
└─$ cd sample_dir

(kali㉿kali)-[~/sample_dir]
└─$ pwd
/home/kali/sample_dir
```

5. 500(r-w)

```
(kali㉿kali)-[~]
└─$ chmod 500 sample.txt

(kali㉿kali)-[~]
└─$ chmod 500 sample_dir

(kali㉿kali)-[~]
└─$ ls -ld sample.txt sample_dir
dr-x----- 2 kali kali 4096 Aug 22 13:15 sample_dir
-r-x----- 1 kali kali    6 Aug 22 13:27 sample.txt

(kali㉿kali)-[~]
└─$ cat sample.txt
Hello

(kali㉿kali)-[~]
└─$ echo "Hii" > sample.txt
zsh: permission denied: sample.txt

(kali㉿kali)-[~]
└─$ ls sample_dir
aanshi.txt

(kali㉿kali)-[~]
└─$ cd s
cd: no such file or directory: s

(kali㉿kali)-[~]
└─$ cd sample_dir

(kali㉿kali)-[~/sample_dir]
└─$ pwd
/home/kali/sample_dir

(kali㉿kali)-[~/sample_dir]
└─$
```

6. 600(rw-)

```
(kali㉿kali)-[~]
└─$ chmod 600 sample.txt

(kali㉿kali)-[~]
└─$ chmod 600 sample_dir

(kali㉿kali)-[~]
└─$ ls -ld sample.txt sample_dir
drw----- 2 kali kali 4096 Aug 22 13:15 sample_dir
-rw----- 1 kali kali    6 Aug 22 13:27 sample.txt

(kali㉿kali)-[~]
└─$ cat sample.txt
Hello

(kali㉿kali)-[~]
└─$ echo "My Name is Aansh" > sample.txt

(kali㉿kali)-[~]
└─$ cat sample.txt
My Name is Aansh

(kali㉿kali)-[~]
└─$ ls sample_dir
ls: cannot access 'sample_dir/aansh1.txt': Permission denied
aansh1.txt

(kali㉿kali)-[~]
└─$ cd sample_dir
cd: permission denied: sample_dir

(kali㉿kali)-[~]
└─$ touch sample_dir/aansh1.txt
touch: cannot touch 'sample_dir/aansh1.txt': Permission denied

(kali㉿kali)-[~]
```


7. 700(rwx)

```
(kali㉿kali)-[~]
└─$ chmod 700 sample.txt

(kali㉿kali)-[~]
└─$ chmod 700 sample_dir

(kali㉿kali)-[~]
└─$ ls -ld sample.txt sample_dir
drwx----- 2 kali kali 4096 Aug 22 13:15 sample_dir
-rwx----- 1 kali kali  17 Aug 22 13:43 sample.txt

(kali㉿kali)-[~]
└─$ cat sample.txt
My Name is Aansh

(kali㉿kali)-[~]
└─$ echo "My Name is Hegde Aansh" > sample.txt

(kali㉿kali)-[~]
└─$ cat sample.txt
My Name is Hegde Aansh

(kali㉿kali)-[~]
└─$ ./sample.txt
./sample.txt: 1: My: not found

(kali㉿kali)-[~]
└─$ cd sample_dir

(kali㉿kali)-[~/sample_dir]
└─$ touch sample_dir/aansh1.txt
touch: cannot touch 'sample_dir/aansh1.txt': No such file or directory

(kali㉿kali)-[~/sample_dir]
└─$ touch aansh1.txt

(kali㉿kali)-[~/sample_dir]
└─$ ls
aansh1.txt

(kali㉿kali)-[~/sample_dir]
└─$ touch hegde1.txt

(kali㉿kali)-[~/sample_dir]
└─$ ls
aansh1.txt  hegde1.txt

(kali㉿kali)-[~/sample_dir]
└─$ rm hegde1.txt

(kali㉿kali)-[~/sample_dir]
└─$ ls
aansh1.txt
```

Permission	Regular File (sample.txt) – Test	Observation	Directory (sample_dir) – Test	Observation
400(r--)	Try to read and modify the file	Will be able to read successfully but can't modify it(permission denied)	Try to list and enter the directory	Will list the filename but can't access it and cannot enter the directory
200(-w-)	Try to read and modify the file	We cannot read the file but can modify it	Try to list, enter,create a file	Cannot list,enter or create a file
100(--x)	Try to read and execute the file	Cannot read or execute the file	Try to list and enter the directory	Cannot list the directory but we can enter it.
500(r-x)	Try to read and modify the file	Will be able to read the file but cannot modify it	Try to list and enter the directory	Can list and enter the directory
600(rw-)	Try to read and modify the file	Can both read and modify the file	Try to list,enter and create a file	Can list filename but cannot enter directory or create files
700(rwx)	Try to read, modify and execute the file	Will be able to read, modify and execute the file	Try to list,enter , create and delete files	Will be able to list,enter, create and delete files

Question:

1. **What does read (r) permission allow you to do with a regular file?**
It allows you to read and view the contents of the file.
2. **What does read (r) permission allow you to do with a directory?**
It allows you to list the names of files and directories inside the directory.
3. **What does write (w) permission mean for a regular file?**
It allows you to modify, overwrite, or change the contents of the file.
4. **What does write (w) permission mean for a directory?**
It allows you to create, delete, and rename files or directories inside it, usually when combined with execute (x) permission.
5. **What does execute (x) permission mean for a regular file?**
It allows you to execute the file as a program or script, if it is a valid executable.
6. **Why is execute (x) permission important for accessing a directory?**
It allows you to enter and traverse the directory and access its contents.
7. **Which permission combination allows the owner to fully access a regular file?**
700 (rwx) — the owner can read, write, and execute the file.
8. **Which permission combination allows the owner to fully access a directory?**
700 (rwx) — the owner can list, enter, create, delete, and manage files in the directory.
9. **Why do the same r, w, and x permissions behave differently for a regular file and a directory?**
The permissions apply to different types of objects. For a regular file, r means reading contents, w means modifying contents, and x means executing the file. For a directory, r means listing its contents, w means creating/deleting/rename entries, and x means entering or traversing the directory.